

Conformal Prediction for Hierarchical Data

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My thesis in buzz words

PhD objective: Probabilistic forecasting for EV smart charging

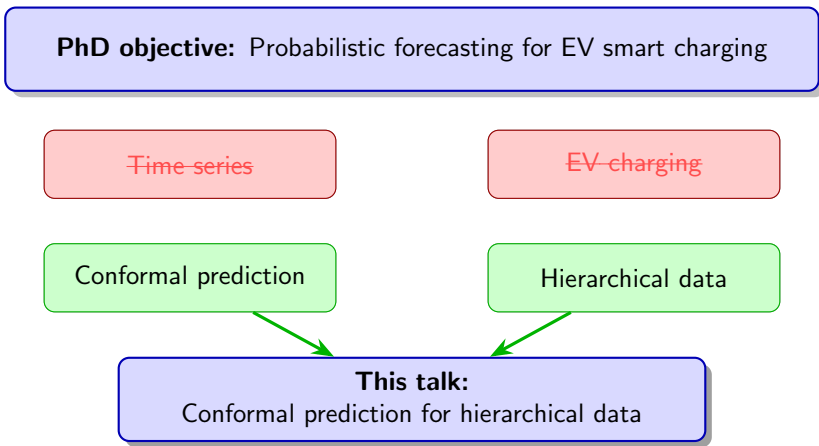
Time series

EV charging

Conformal prediction

Hierarchical data

My thesis in buzz words

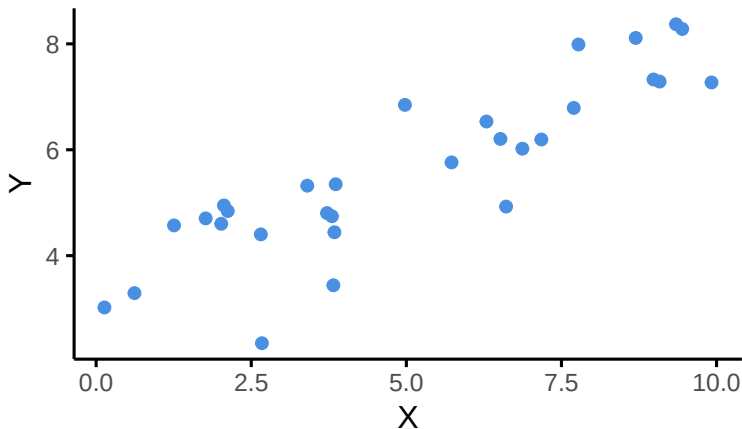


Outline

- 1. Split Conformal Prediction (SCP) in the univariate setting**
2. Conformal Prediction for Multivariate Data
3. Conformal Prediction for Hierarchical Data
4. Conclusion

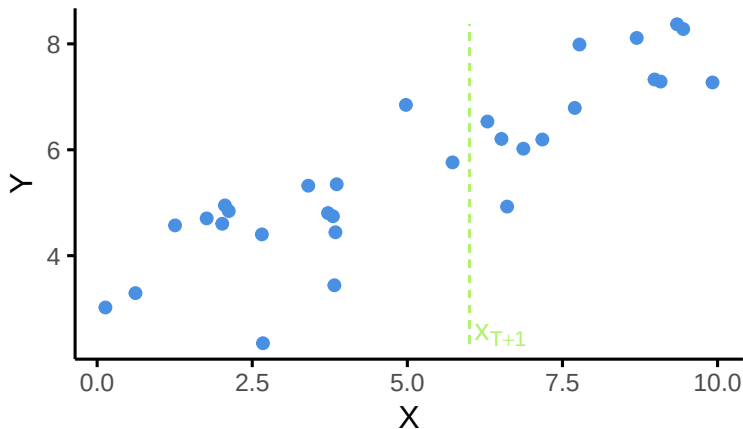
Setting – univariate

Regression setting with covariates $\mathbf{x}_t \in \mathbb{R}^d$ and targets $y_t \in \mathbb{R}$. Observations $(\mathbf{x}_t, y_t)_{t \in [T]}$ and covariate $\mathbf{x}_{T+1}$ are available.



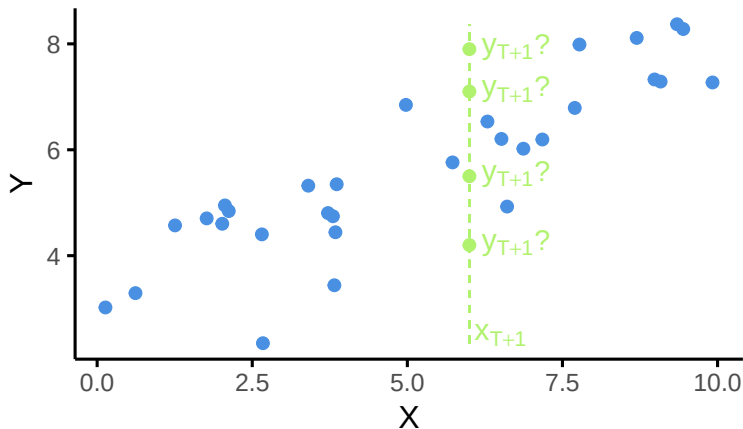
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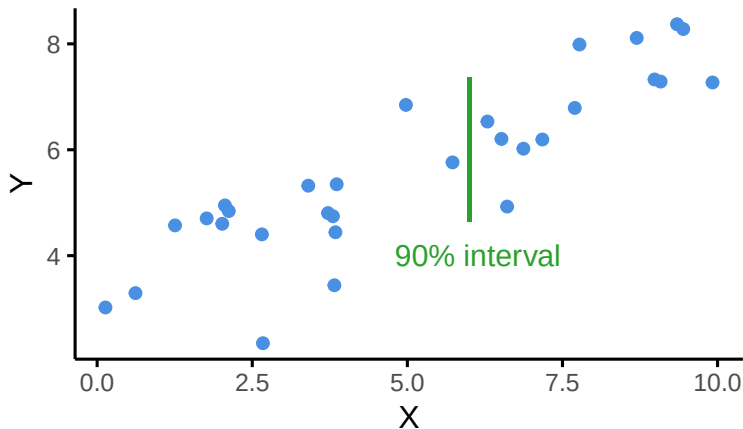
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Setting – univariate

Regression setting with covariates $\mathbf{x}_t \in \mathbb{R}^d$ and targets $y_t \in \mathbb{R}$. Observations $(\mathbf{x}_t, y_t)_{t \in [T]}$ and covariate $\mathbf{x}_{T+1}$ are available.



Coverage guarantee of SCP – univariate

SCP¹ achieves the following objective

Objective: build a prediction set $C(\mathbf{x}_{T+1})$ such that:

Theorem 1: Coverage of SCP [Vovk et al. (2005)]

If $(\mathbf{x}_t, y_t)_{t \in [T+1]}$ are i.i.d. (or exchangeable), then

$$\mathbb{P}(y_{T+1} \in C(\mathbf{x}_{T+1})) \geq 1 - \alpha.$$

Moreover, under an additional continuity assumption

$$\mathbb{P}(y_{T+1} \in C(\mathbf{x}_{T+1})) \leq 1 - \alpha + \mathcal{O}\left(\frac{1}{T}\right).$$

¹the name SCP comes from Lei et al. (2018) and is a rebranding of inductive conformal prediction.

Illustration of SCP with absolute residuals – univariate

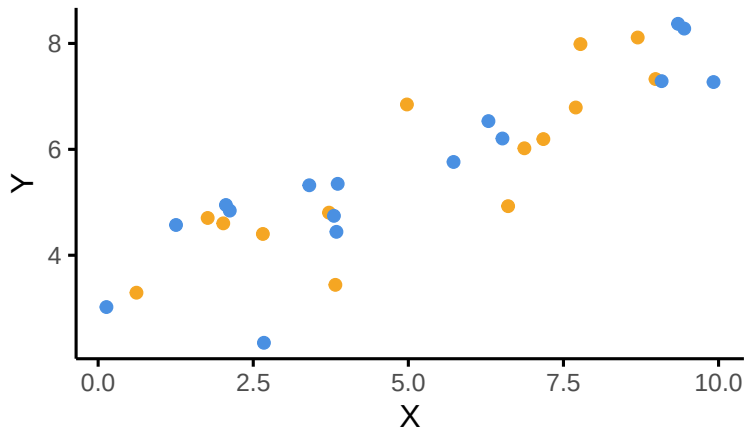


Illustration of SCP with absolute residuals – univariate

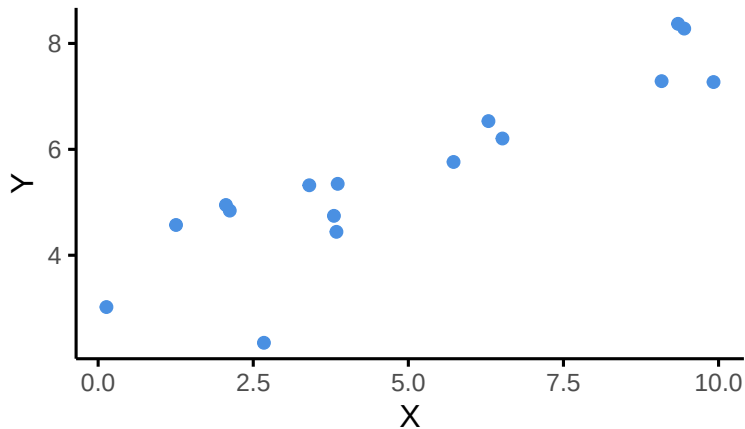


Illustration of SCP with absolute residuals – univariate

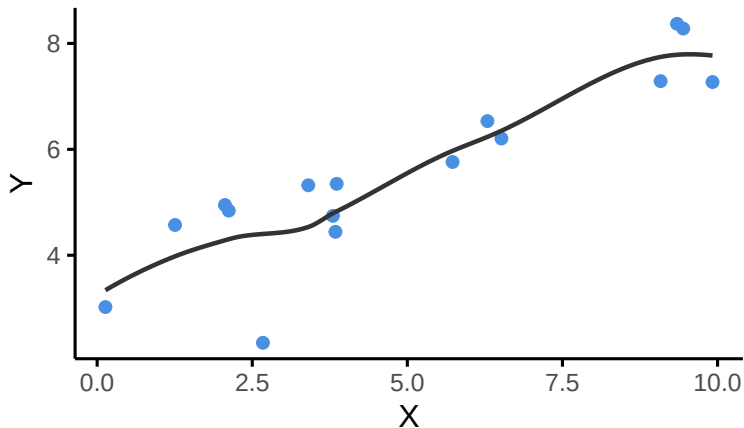


Illustration of SCP with absolute residuals – univariate

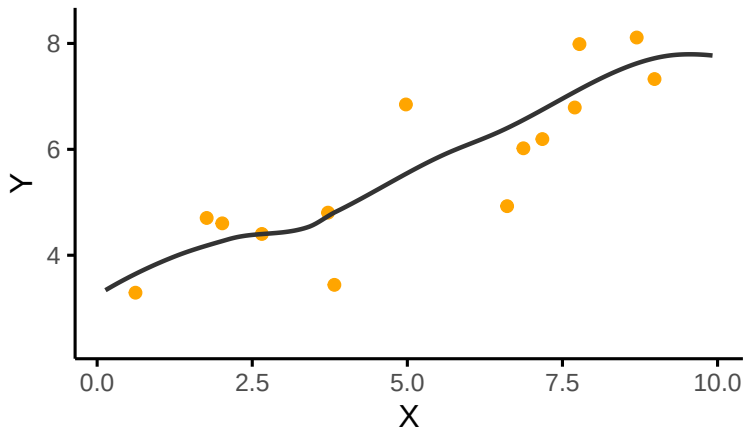


Illustration of SCP with absolute residuals – univariate

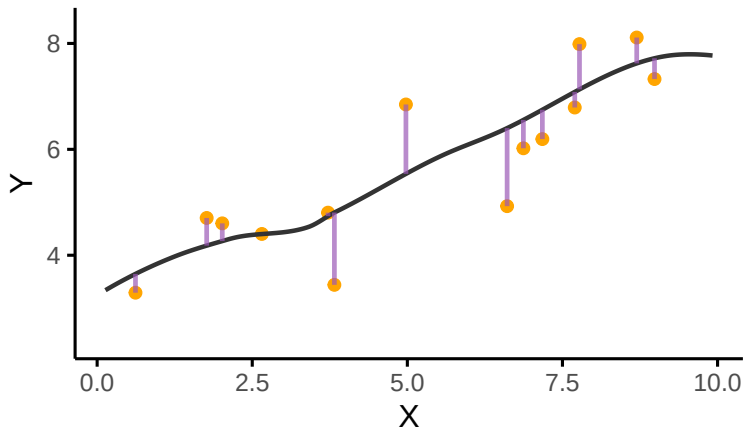


Illustration of SCP with absolute residuals – univariate

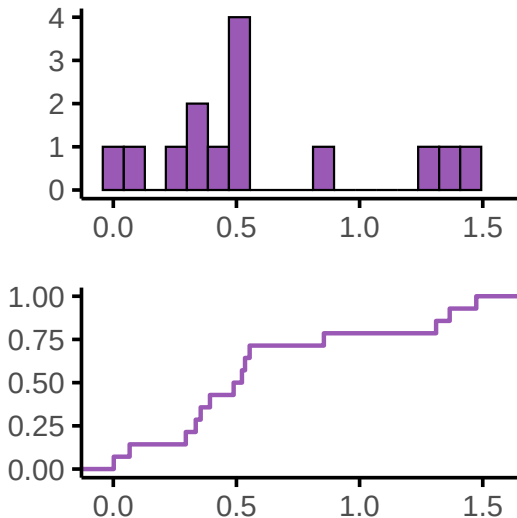


Illustration of SCP with absolute residuals – univariate

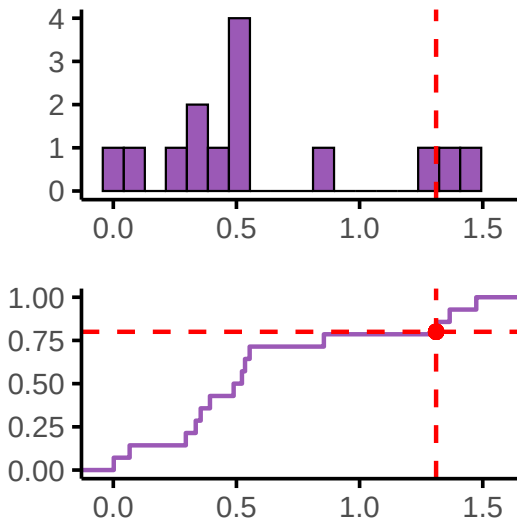
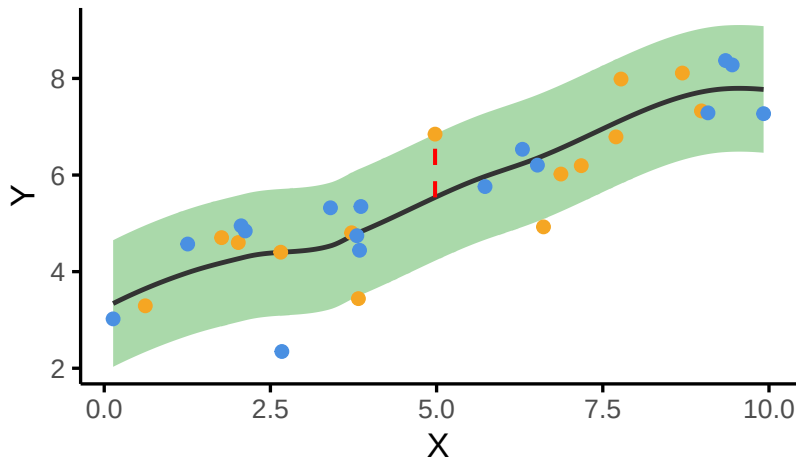
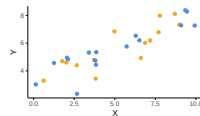


Illustration of SCP with absolute residuals – univariate



Summary of SCP – univariate

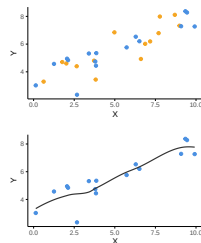
Split the data between $\mathcal{D}_{\text{train}}$ and $\mathcal{D}_{\text{calib}}$



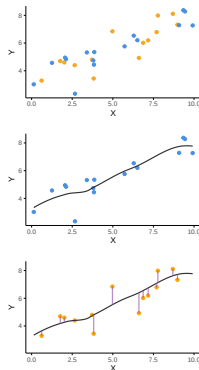
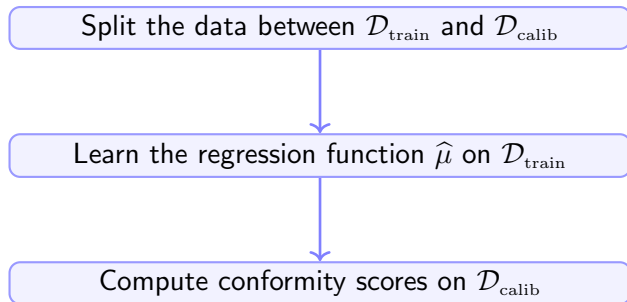
Summary of SCP – univariate

Split the data between $\mathcal{D}_{\text{train}}$ and $\mathcal{D}_{\text{calib}}$

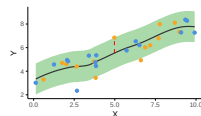
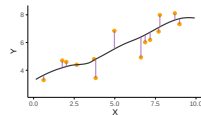
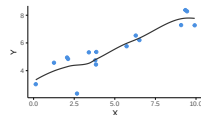
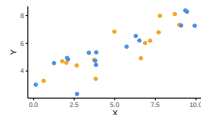
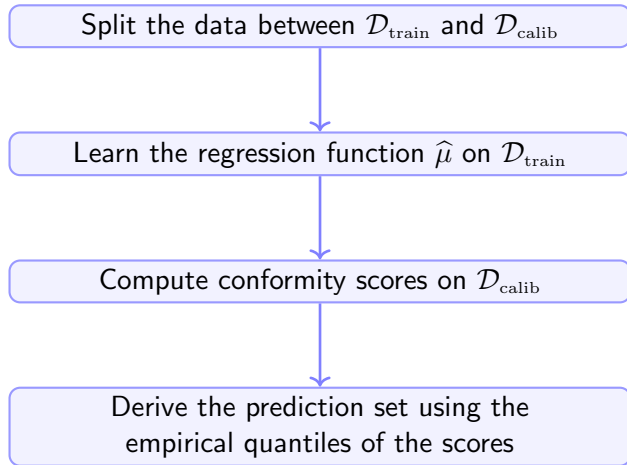
Learn the regression function $\hat{\mu}$ on $\mathcal{D}_{\text{train}}$



Summary of SCP – univariate



Summary of SCP – univariate



Extension of SCP to signed residuals [Linusson et al. (2014)]

Slight modification of SCP (e.g. to deal with biased predictors): consider signed residuals as conformity scores.

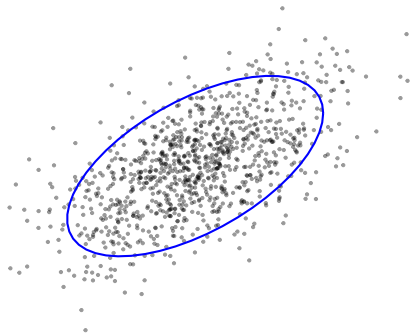
- learn the regressor $\hat{\mu}(\cdot) = \mathcal{A}((\mathbf{x}_t, y_t)_{t \in \mathcal{D}_{\text{train}}})$
- for $t \in \mathcal{D}_{\text{calib}}$, let $\hat{y}_t = \hat{\mu}(\mathbf{x}_t)$ and compute the conformity score $\hat{s}_t = y_t - \hat{y}_t$
- order the conformity scores: $\hat{s}_{(1)} \leq \dots \leq \hat{s}_{(T_{\text{calib}})}$
- set and return $\hat{C}(\cdot) = \left[\hat{\mu}(\cdot) + \hat{s}_{(\lfloor (T_{\text{calib}}+1)\alpha/2 \rfloor)}, \hat{\mu}(\cdot) + \hat{s}_{(\lceil (T_{\text{calib}}+1)(1-\alpha/2) \rceil)} \right]$

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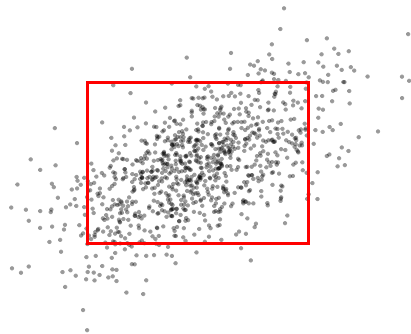
Prediction Regions for Multivariate Data: Validity

In regression for multivariate targets $\mathbf{y} \in \mathbb{R}^m$ and covariates $\mathbf{x} \in \mathbb{R}^d$ with $m > 1$, we can consider 2 types of prediction regions:



which targets joint coverage:

$$\mathbb{P}(\mathbf{y}_{T+1} \in E(\mathbf{x}_{T+1})) \approx 1 - \alpha$$



which targets component-wise coverage:

$$\forall i \in [m], \quad \mathbb{P}(y_{T+1,i} \in C_i(\mathbf{x}_{T+1})) \approx 1 - \alpha$$

Ellipsoidal SCP (joint-coverage)

Challenge: build a real-valued score function that takes dependencies into account.

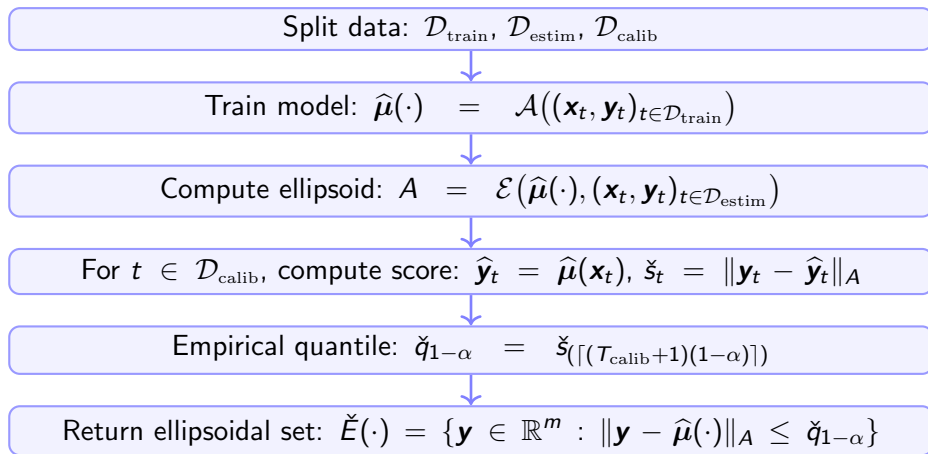
A flexible method consists in ellipsoidal conformal sets (Johnstone and Cox, 2021).

$$\check{s}_t = \|\mathbf{y}_t - \hat{\mathbf{y}}_t\|_A \stackrel{\text{def}}{=} \sqrt{(\mathbf{y}_t - \hat{\mathbf{y}}_t)^\top A (\mathbf{y}_t - \hat{\mathbf{y}}_t)},$$

with $\hat{\mathbf{y}}_t = \hat{\boldsymbol{\mu}}(\mathbf{x}_t)$ and matrix $A \succeq 0$.

Remark: if $A = \Sigma^{-1}$, the score is the *Mahalanobis* distance.

Ellipsoidal SCP: Algorithm 1 (baseline joint-coverage)



Component-wise SCP

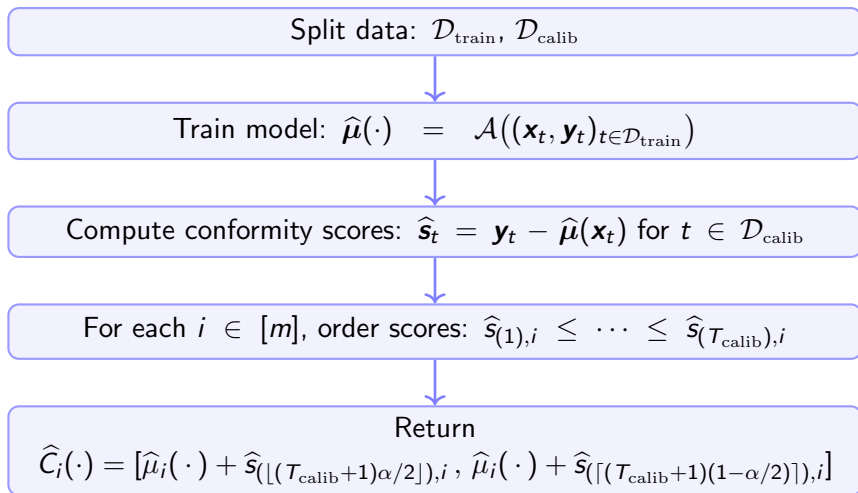
Underexplored objective, which appears naturally in industrial applications—with vectorial target under individual scrutiny.

Example: Load consumption on the twelve french regions estimated jointly, for which uncertainty should be component by component.

This leads to the natural component-wise coverage objective:

$$\forall i \in [m], \quad \mathbb{P}(y_{T+1,i} \in C_i(\mathbf{x}_{T+1})) \approx 1 - \alpha.$$

Component-wise SCP: Algorithm 3 (baseline component-wise)



Immediate Results for the baselines (informal)

If the $(\mathbf{x}_t, \mathbf{y}_t)_{t \in [T+1]}$ are i.i.d., $\check{E}(\mathbf{x}_{T+1})$ from Alg.1 achieves joint coverage:

$$\mathbb{P}(\mathbf{y}_{T+1} \in \check{E}(\mathbf{x}_{T+1})) \approx 1 - \alpha,$$

and the $\hat{C}_i(\mathbf{x}_{T+1})$ from Alg.3 achieve component-wise coverage:

$$\forall i \in [m], \quad \mathbb{P}(\mathbf{y}_{T+1,i} \in \hat{C}_i(\mathbf{x}_{T+1})) \approx 1 - \alpha.$$

Our Contributions

If $\mathbf{y}_{T+1,i}$ follow some hierarchical structure, adding a projection step improves efficiency.

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Hierarchical Data

Denote $y_{B_1}, \dots, y_{B_n}$ the n first components of $\mathbf{y} \in \mathbb{R}^m$. The target $\mathbf{y}$ is hierarchical if

$$\mathbf{y} = H\mathbf{y}_{B_{1:n}} \quad \text{with for example} \quad H = \begin{bmatrix} \text{Id}_2 \\ 1 & 1 \end{bmatrix} \quad \text{which corresponds to} \quad y_{\text{Tot}} = y_{B_1} + y_{B_2}.$$

Example: Consider $\mathbf{y}_{B_{1:12}}$, the load consumption within the twelve french regions and y_{Tot} the french one. Then, $\mathbf{y}$ is hierarchical since $y_{\text{Tot}} = y_{B_1} + \dots + y_{B_{12}}$.

Coherent Forecast

Problem: In general, $\hat{\mathbf{y}} = \hat{\boldsymbol{\mu}}(\mathbf{x})$ does not satisfy the linear constraint.

How can we effectively leverage this structural information?

Bottom-Up: Trust $\hat{y}_{B_1}$ and $\hat{y}_{B_2}$ to build $\hat{y}_{\text{Tot}} = \hat{y}_{B_1} + \hat{y}_{B_2}$

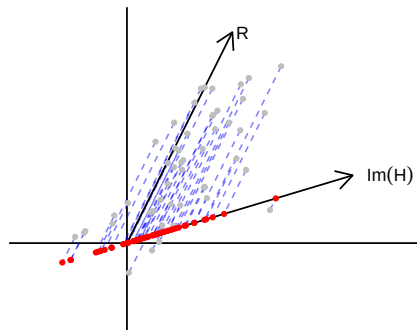
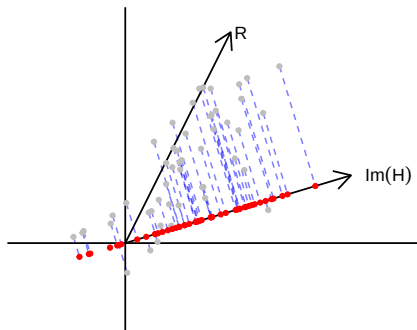
Top-Down: Trust $\hat{y}_{\text{Tot}}$ to build $\tilde{y}_{B_1} = \frac{\hat{y}_{B_1}}{\hat{y}_{B_1} + \hat{y}_{B_2}} \hat{y}_{\text{Tot}}$ and $\tilde{y}_{B_2} = \frac{\hat{y}_{B_2}}{\hat{y}_{B_1} + \hat{y}_{B_2}} \hat{y}_{\text{Tot}}$.

Forecast Reconciliation

Hyndman et al. (2011) consider projection onto the coherent subspace $\text{Im}(H)$:

$$\tilde{\mathbf{y}} = P\hat{\mathbf{y}}.$$

For example,



Forecast Reconciliation: how to choose the projection matrix?

Several methods developed for point forecasting (e.g. Hyndman et al., 2011; Wickramasuriya et al., 2019; Panagiotelis et al., 2021).

OLS: basic orthogonal projection

$$P_{\text{OLS}} = H(H^{\top}H)^{-1}H^{\top}.$$

WLS: takes into account the confidence on each individual forecaster

$$P_{\text{WLS}} = H(H^{\top}\text{Diag}(\Sigma)^{-1}H)^{-1}H^{\top}\text{Diag}(\Sigma)^{-1}.$$

MinT: takes into account the confidence AND dependencies within forecasters

$$P_{\text{MinT}} = H(H^{\top}\Sigma^{-1}H)^{-1}H^{\top}\Sigma^{-1}.$$

Bonus Califrais: objective for point forecasting

If $\hat{\mathbf{y}}$ unbiased, or equivalently $\hat{\mathbf{s}} = \mathbf{y} - \hat{\mathbf{y}}$ centered, the multivariate extension of MSE is

$$\text{MSE}(\hat{\mathbf{y}}, \mathbf{y}) = \mathbb{E}[\|\hat{\mathbf{s}}\|^2] = \mathbb{E}[\hat{\mathbf{s}}^\top \hat{\mathbf{s}}] = \mathbb{E}[\text{Tr}(\hat{\mathbf{s}} \hat{\mathbf{s}}^\top)] = \text{Tr}(\mathbb{E}[\hat{\mathbf{s}} \hat{\mathbf{s}}^\top]) = \text{Tr}(\Sigma).$$

Similarly, for weighted MSE,

$$\text{MSE}(\hat{\mathbf{y}}, \mathbf{y}, W) = \mathbb{E}[\|\hat{\mathbf{s}}\|_W^2] = \text{Tr}(W\Sigma).$$

By linearity of a projection, $\tilde{\mathbf{y}}$ are also unbiased and since $\mathbf{y} - P\hat{\mathbf{y}} = P(\mathbf{y} - \hat{\mathbf{y}}) = \tilde{\mathbf{s}}$,

$$\text{MSE}(\tilde{\mathbf{y}}, \mathbf{y}, W) = \mathbb{E}[\text{Tr}(W \tilde{\mathbf{s}} \tilde{\mathbf{s}}^\top)] = \text{Tr}(WP\Sigma P^\top).$$

Thus the objective of minimizing $\text{Tr}(WP\Sigma P^\top)$, that induces the name MinT.

Reconciled Ellipsoidal SCP: idea (joint-coverage)

Baselines achieve coverage, still we can improve the efficiency of the prediction regions.

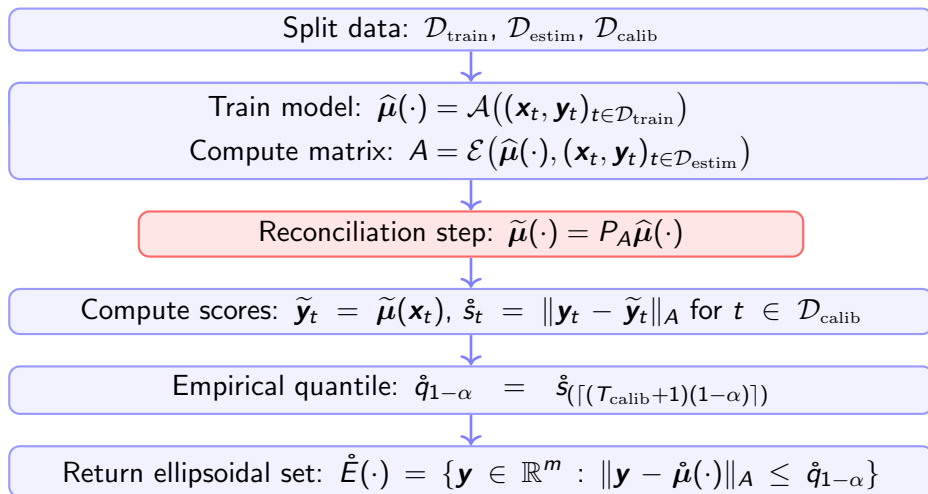
For joint coverage, it makes sense to look for minimum volume prediction regions i.e.

$$\text{minimize} \quad \mathbb{E} \left[\mathfrak{L}_m(E(\mathbf{x}_{T+1})) \right].$$

By a Pythagorean theorem, $P_A = H(H^\top AH)^{-1}H^\top A$ satisfies

$$\|\mathbf{y} - P_A \hat{\mathbf{y}}\|_A \leq \|\mathbf{y} - \hat{\mathbf{y}}\|_A.$$

Reconciled Ellipsoidal SCP: Algorithm 2 (joint-coverage)



Reconciled Ellipsoidal SCP: immediate result

Denote $\check{E}(\mathbf{x}_{T+1})$ and $\mathring{E}(\mathbf{x}_{T+1})$, the ellipsoidal conformal regions of Alg.1 resp. Alg.2.

Under the i.i.d. assumption,

$$\mathbb{P}(\mathbf{y}_{T+1} \in \check{E}(\mathbf{x}_{T+1})) \approx 1 - \alpha \quad \text{and} \quad \mathbb{P}(\mathbf{y}_{T+1} \in \mathring{E}(\mathbf{x}_{T+1})) \approx 1 - \alpha .$$

While for all $\mathbf{x} \in \mathbb{R}^d$,

$$\mathfrak{L}_m(\mathring{E}(\mathbf{x})) \leq \mathfrak{L}_m(\check{E}(\mathbf{x})) .$$

How to define efficiency for component-wise SCP?

Weak version:

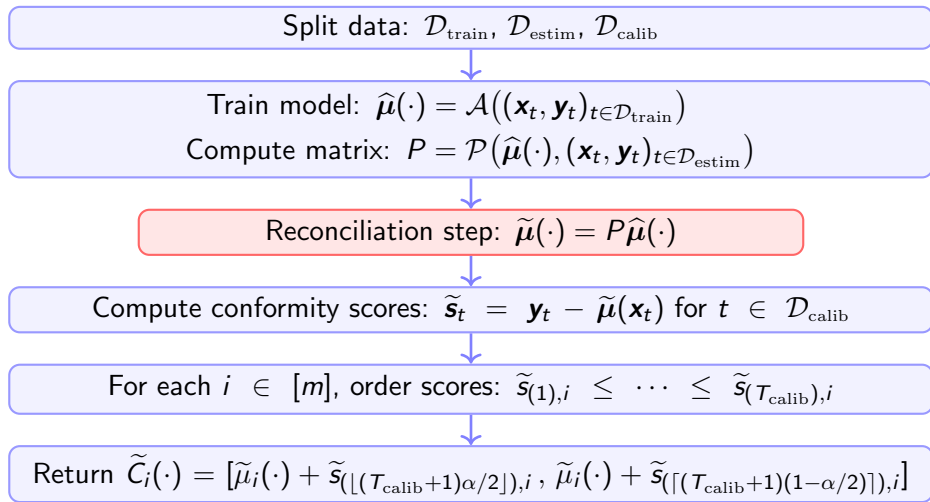
$$\text{minimize} \quad \mathbb{E} \left[\sum_{i=1}^m w_i \mathcal{L}_1(C_i(\mathbf{x}_{T+1}))^2 \right],$$

with $\mathbf{w} = (w_1, \dots, w_m)^\top$ a positive weight vector.

Strong version:

$$\forall i \in [m], \quad \text{minimize} \quad \mathbb{E} \left[\mathcal{L}_1(C_i(\mathbf{x}_{T+1}))^2 \right].$$

Reconciled Component-wise SCP: Algorithm 4



Results (component-wise coverage)

Denote $\hat{C}(\mathbf{x}_{T+1})$ and $\tilde{C}(\mathbf{x}_{T+1})$, the rectangular conformal regions of Alg.3 resp. Alg.4.

Under the i.i.d. assumption, for all $i \in [m]$,

$$\mathbb{P}(y_{T+1,i} \in \tilde{C}_i(\mathbf{x}_{T+1})) \approx 1 - \alpha \quad \text{and} \quad \mathbb{P}(y_{T+1,i} \in \hat{C}_i(\mathbf{x}_{T+1})) \approx 1 - \alpha.$$

But no immediate efficiency result (e.g. we don't have $\tilde{s} \leq \hat{s}$).

Additional Assumptions for efficiency results

Disclaimer:

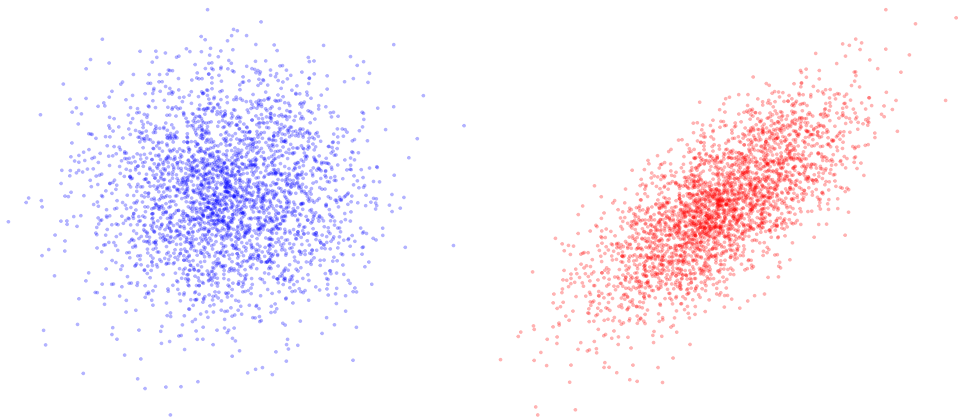
SCP is a **distribution-free** framework that gives **finite-sample** coverage guarantees under mild assumptions.

When it comes to efficiency, additional assumptions are **always** required.

Assumption 1 (weak version): The (i.i.d.) conformity scores $\hat{\mathbf{s}}_t = \mathbf{y}_t - \hat{\mathbf{y}}_t$ follow some elliptical distribution (parameters are unknown).

Assumption 2 (strong version): The (i.i.d.) conformity scores $\hat{\mathbf{s}}_t = \mathbf{y}_t - \hat{\mathbf{y}}_t$ follow some elliptical distribution (the variance $\Sigma \succ 0$ exists and is known).

A word on elliptical distribution



Examples: multivariate normal distributions, t -distributions and uniform distribution on an ellipse.

Efficiency results (component-wise SCP)

For any positive vector $\mathbf{w}$, if Alg.4 uses $P = H(H^\top \text{diag}(\mathbf{w})H)^{-1}H^\top \text{diag}(\mathbf{w})$ then

Theorem 2: Efficiency – weak version [Principato et al. (2024)]

Under i.i.d. elliptical scores,

$$\mathbb{E} \left[\sum_{i=1}^m w_i \mathfrak{L}_1(\tilde{C}_i(\mathbf{x}_{T+1}))^2 \right] \leq \mathbb{E} \left[\sum_{i=1}^m w_i \mathfrak{L}_1(\hat{C}_i(\mathbf{x}_{T+1}))^2 \right].$$

Remark: The practitioner chooses $\mathbf{w}$ to select components for which he does not want to be less efficient.

Efficiency results (component-wise SCP)

If Alg.4 uses $P_{\Sigma^{-1}} = H(H^\top \Sigma^{-1} H)^{-1} H^\top \Sigma^{-1}$, then it outperforms Alg.4 with any other choice of projection matrix P .

Theorem 3: Efficiency – strong version [Principato et al. (2024)]

Under i.i.d. elliptical scores (with known variance $\Sigma \succ 0$),

$$\forall i \in [m], \quad \mathbb{E} \left[\mathfrak{L}_1(\tilde{C}_i^*(\mathbf{x}_{T+1}))^2 \right] \leq \mathbb{E} \left[\mathfrak{L}_1(\tilde{C}_i(\mathbf{x}_{T+1}))^2 \right].$$

In particular, we also have

$$\forall i \in [m], \quad \mathbb{E} \left[\mathfrak{L}_1(\tilde{C}_i^*(\mathbf{x}_{T+1}))^2 \right] \leq \mathbb{E} \left[\mathfrak{L}_1(\hat{C}_i(\mathbf{x}_{T+1}))^2 \right].$$

Practical assessment (Ellipsoidal SCP)

Design of the experiment:

- 6 hierarchical structures from the smallest (16 nodes) to the largest (1 801 nodes)
- nominal coverage $1 - \alpha = 90\%$
- 10^6 observations and 1000 runs for robustness

For joint-coverage, we get valid regions with

Matrix A	Config.	Alg.1	Alg.2
$\hat{\Sigma}^{-1}$	1	17.4 ± 0.6	16.5 ± 0.55
	2	3.36 ± 0.12	3.19 ± 0.11

Table: empirical averages of normalized volumes $\pm \sqrt{1.96 \times \text{standard errors}}$.

Practical assessment (Component-wise SCP)

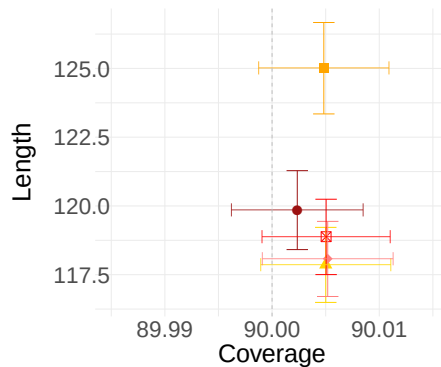
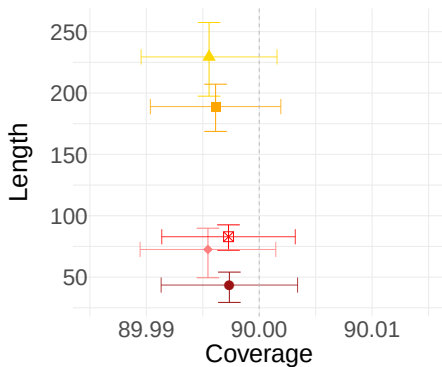
For component-wise coverage, we get valid regions with

Config.	Alg.3-Direct	Alg.4-OLS	Alg.4-WLS	Alg.4-MinT
1	876 $\pm$ 254	787 $\pm$ 226	322 $\pm$ 131	216 $\pm$ 47
2	871 $\pm$ 253	753 $\pm$ 216	308 $\pm$ 116	246 $\pm$ 51
3	3032 $\pm$ 467	2954 $\pm$ 455	1869 $\pm$ 377	1502 $\pm$ 578
4	3036 $\pm$ 479	2901 $\pm$ 458	1581 $\pm$ 340	1404 $\pm$ 571
5	10424 $\pm$ 885	10358 $\pm$ 880	8861 $\pm$ 853	10613 $\pm$ 918
6	10621 $\pm$ 889	10460 $\pm$ 875	7673 $\pm$ 785	10503 $\pm$ 905

Table: root-empirical averages of total squared length $\pm \sqrt{1.96 \times \text{standard errors}}$.

Practical assessment (Component-wise SCP)

Methods:  Direct  OLS  WLS  Combi  MinT



Toolbox to prove the efficiency results

- **Tool 1:** Stability under linear transform of elliptical distribution.
- **Tool 2:** Design of the scores $P\hat{\mathbf{s}}_t = \mathbf{y}_t - P\hat{\mathbf{y}}_t$.
- **Tool 3:** Connection with Trace inequalities from Forecast Reconciliation.

Lemma 1 (MinT) [Wickramasuriya et al. (2019)]

Let $W \succeq 0$ and $\Sigma \succ 0$. Then, for all projection matrices P onto $\text{Im}(H)$,

$$\text{Tr}(WP_{\Sigma^{-1}}\Sigma P_{\Sigma^{-1}}^{\top}) \leq \text{Tr}(WP\Sigma P^{\top}), \quad \text{where} \quad P_{\Sigma^{-1}} = H(H^{\top}\Sigma^{-1}H)^{-1}H^{\top}\Sigma^{-1}.$$

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Conclusion and future works

On going work: Use case on EV charging – including extension to time series and hierarchies that vary over time.

Recent Preprint (Nov 2025): Conformal prediction beyond exchangeability, using Blackwell's Approachability to handle both coverage and efficiency.

We derive an explicit algorithm for which we ensure coverage and stronger theoretical efficiency guarantees than existing methods.

References

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Appendix: Sketch of proof of an important trace inequality

Lemma 2 [Principato et al. (2024) inspired by Panagiotelis et al. (2021)]

Consider P_W , the projection onto $\text{Im}(H)$ in $\|\cdot\|_W$ -norm with $W \succ 0$. Then $\forall M \in \mathbb{R}^{m \times k}$,

$$0 \leq \text{Tr}(WP_W MM^\top P_W^\top) \leq \text{Tr}(WMM^\top).$$

Proof. Set $\mathbf{z}' \sim \mathcal{N}(\mathbf{0}, \text{Id}_k)$; by a Pythagorean theorem, $\|P_W M \mathbf{z}'\|_W^2 \leq \|M \mathbf{z}'\|_W^2$ a.s.

$$\begin{aligned} \mathbb{E}[\|P_W M \mathbf{z}'\|_W^2] &= \mathbb{E}[(P_W M \mathbf{z}')^\top W P_W M \mathbf{z}'] \\ &= \text{Tr}\left(WP_W M \mathbb{E}[\mathbf{z}'(\mathbf{z}')^\top] M^\top P_W^\top\right) = \text{Tr}\left(WP_W M M^\top P_W^\top\right). \end{aligned}$$

Similarly, $\mathbb{E}[\|M \mathbf{z}'\|_W^2] = \text{Tr}(WMM^\top)$, which concludes using above inequality.